

ORYAHOVO, Bulgaria

WMO: 155140

Lat: 43.683N

Lon: 23.967E

Elev: 102

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	17.3	9.9	9.2	13.9	14.4	11.5	18.6	44.2	37.4	36.6	37.5	5.1	90	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	92.7	72.9	89.6	72.1	87.1	71.0	75.3	88.2	73.6	86.2	72.1	84.4	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	115.2	83.1	69.3	108.2	81.0	67.8	102.4	79.3	38.8	88.1	37.2	86.4	35.8	84.6	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.6	35.1	26.2	DB	8.7	98.9	6.0	4.5	4.4	102.1	0.9	104.7	-2.4	107.2	-6.8	110.5	(4)
(5)			WB	8.5	81.3	5.4	3.3	4.6	83.7	1.5	85.6	-1.5	87.5	-5.4	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	31.6	36.3	46.0	55.9	66.1	72.1	76.9	76.1	67.6	55.6	45.0	34.2		
	DBStd	17.34	8.98	9.70	8.31	5.87	5.90	5.73	4.94	4.98	6.03	7.15	6.92	7.67		
	HDD50	1856	572	392	174	16	0	0	0	0	0	28	182	492		
	HDD65	4682	1036	805	589	279	60	10	1	2	43	301	601	955		
	CDD50	3822	2	7	50	193	500	662	835	810	527	203	31	2		
	CDD65	1172	0	0	0	5	96	222	371	347	120	11	0	0		
	CDH74	9457	0	0	2	34	566	1720	3373	2958	750	51	3	0		
	CDH80	3360	0	0	0	4	99	539	1369	1147	197	5	0	0		
(14) Wind	WSAvg	6.5	6.4	7.6	8.3	6.6	6.3	6.6	6.5	5.8	6.2	5.7	6.1	6.3		
(15) Precipitation	PrecAvg	22.1	1.5	1.3	1.6	1.9	2.6	2.4	2.2	1.5	2.0	2.0	1.4	1.7		
	PrecMax	36.8	3.5	2.8	3.7	4.8	4.5	4.8	4.8	7.7	5.9	4.3	3.9	4.1		
	PrecMin	12.9	0.3	0.1	0.4	0.1	0.6	0.5	0.4	0.2	0.2	0.2	0.0	0.1		
	PrecStd	5.4	0.9	0.7	0.9	1.0	1.2	1.1	1.4	1.5	1.5	1.3	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	62.8	71.0	78.5	86.5	93.6	98.5	96.6	89.6	79.3	68.7	58.8		
		MCWB	48.4	51.4	54.6	61.9	68.3	72.8	73.4	72.6	70.0	66.1	57.5	50.1		
	2%	DB	52.6	58.6	66.4	72.4	82.4	88.8	93.4	91.9	85.2	74.0	62.7	51.2		
		MCWB	45.6	48.7	52.4	58.8	66.6	71.9	73.3	72.6	67.9	62.8	55.0	46.2		
	5%	DB	48.2	54.1	62.4	68.8	79.7	85.9	90.1	89.1	81.4	69.9	58.5	46.2		
		MCWB	42.9	46.3	50.4	56.8	65.3	70.8	72.6	71.8	66.6	61.0	53.0	43.1		
	10%	DB	43.7	49.9	58.8	66.1	76.9	83.0	87.2	86.4	78.0	66.3	55.0	43.7		
		MCWB	40.1	43.7	48.8	55.2	64.0	69.5	71.9	70.3	65.0	58.8	50.9	41.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	52.2	56.4	63.5	70.3	76.1	78.3	76.8	71.4	67.9	59.2	50.7		
		MCDB	56.7	61.7	67.5	75.3	82.8	87.1	89.6	89.8	86.6	77.2	66.5	57.7		
	2%	WB	45.9	49.3	53.6	60.2	68.3	73.7	75.8	74.5	69.3	64.2	55.9	46.7		
		MCDB	51.7	57.2	63.9	70.2	79.6	86.5	88.4	87.9	82.2	72.3	60.4	50.5		
	5%	WB	43.3	46.6	51.3	58.1	66.6	72.0	74.1	72.8	67.6	61.8	54.0	44.0		
		MCDB	47.8	52.9	61.3	66.8	77.5	83.7	86.6	85.5	79.3	68.5	58.0	45.9		
	10%	WB	40.3	44.4	49.3	56.3	65.0	70.5	72.4	71.4	65.8	59.4	51.7	41.4		
		MCDB	43.6	49.4	57.9	64.2	74.8	81.2	84.7	84.1	76.3	65.3	54.6	43.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.7	14.3	15.6	16.4	16.2	17.5	17.7	16.8	13.5	9.5	7.6		
		MCDBR	14.8	18.4	21.6	20.6	20.7	20.6	21.9	21.8	22.4	19.0	16.1	11.8		
		MCWBR	10.5	11.7	11.5	10.3	9.4	8.9	9.8	8.5	10.0	10.5	9.9	8.6		
		MCDBR	13.7	17.3	20.5	18.6	18.7	18.8	19.5	19.6	20.6	17.6	13.1	10.9		
	5% WB	MCWBR	10.0	11.4	11.6	10.1	9.7	9.5	10.0	8.9	10.1	10.7	8.5	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.375	0.415	0.474	0.453	0.453	0.464	0.479	0.446	0.404	0.371	0.348		
	taud		2.294	2.216	2.161	2.013	2.146	2.186	2.159	2.114	2.172	2.278	2.325	2.318		
	Ebn at Noon		241	253	258	251	261	261	257	247	245	240	230	227		
	Edh at Noon		27	35	42	53	47	46	46	47	41	33	27	24		
(44) All-Sky Solar Radiation	RadAvg		460	725	1087	1489	1824	2028	2056	1830	1326	860	491	379		
	RadStd		70	99	117	159	154	140	127	134	105	111	69	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+0.88	N/A	N/A	N/A	N/A	+0.72	N/A	-254	+126	N/A

Nomenclature: See separate page